MEDIUM · PROBLEM-SOLVING AND DATA ANALYSIS · ANSWER KEY

Problem-Solving and Data Analysis

12

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

A

A. $1.43w$

Choice A is correct. The result of increasing a positive quantity w by $x\%$ can be represented by the expression $1+\frac{x}{100}w$. Therefore, the result of increasing a positive quantity w by 43% can be found by substituting 43 for x in the expression $1+\frac{x}{100}w$, which gives $1+\frac{43}{100}w$, or $1.43w$. Thus, the expression $1.43w$ represents the result of increasing a positive quantity w by 43%.

Choice B is incorrect. This is the result of decreasing a positive quantity w by 43%.

Choice C is incorrect. This is the result of increasing a positive quantity w by $4{,}200\%$.

Choice D is incorrect. This is the result of decreasing a positive quantity w by 57%.

Q2

C

C. -0.7

Choice C is correct. A line of best fit is shown in the scatterplot such that as the value of x increases, the value of y decreases. It follows that the slope of the line of best fit shown is negative. The slope of a line in the xy-plane that passes through the points x_1, y_1 and x_2, y_2 can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points $0, 8$ and $10, 1$. Substituting $0, 8$ for x_1, y_1 and $10, 1$ for x_2, y_2 in $\frac{y_2 - y_1}{x_2 - x_1}$ yields the slope of the line being approximately $\frac{1 - 8}{10 - 0}$, which is equivalent to $\frac{-7}{10}$, or -0.7. Therefore, of the given choices, -0.7 is the closest to the slope of this line of best fit.

Choice A is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Choice B is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3

25

The correct answer is 25.4. The average speed is the total distance divided by the total time. The total distance is 11 miles and the total time is 26 minutes. Thus, the average speed is $\frac{11}{26}$ miles per minute. The question asks for the average speed in miles per hour, and there are 60 minutes in an hour; converting miles per minute to miles per hour gives the following:

$$\text{Average speed} = \frac{11 \text{ miles}}{26 \text{ minutes}} \times \frac{60 \text{ minutes}}{1 \text{ hour}}$$

$$= \frac{660}{26} \text{ miles per hour}$$

$$\approx 25.38 \text{ miles per hour}$$

Therefore, to the nearest tenth of a mile per hour, the average speed of Paul Revere's ride would have been 25.4 miles per hour. Note that 25.4 and 127/5 are examples of ways to enter a correct answer.

Q4

C

C. $y = 3.30x + 0.82$

Choice C is correct. A line that most closely fits the data is a line with an approximately balanced number of data points above and below the line. Fitting a line to the data shown results in a line with an approximate slope of 3 and a y-intercept near the point $(0,1)$. An equation for the line can be written in slope-intercept form, $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept. The equation $y = 3.30x + 0.82$ in choice C fits the data most closely.

Choices A and B are incorrect because the slope of the lines of these equations is 0.82, which is a value that is too small to be the slope of the line that fits the data shown. Choice D is incorrect. The graph of this equation has a y-intercept at $(0, -3.30)$, not $(0, 0.82)$. This line would lie below all of the data points, and therefore would not closely fit the data.

Q5

35728

The correct answer is 35,728. It's given that 1 mile = 1,760 yards. It follows that an average speed of 20.300 miles per hour is equivalent to $\frac{20.300 \text{ miles}}{1 \text{ hour}}\frac{1{,}760 \text{ yards}}{1 \text{ mile}}$, or 35,728 yards per hour.

Q6

1015

The correct answer is 1,015. It’s given that the first moon orbits the planet every 252 days. Therefore, it takes the first moon 25229, or 7,308, days to orbit the planet 29 times. It’s also given that the second moon orbits the planet every 287 days. Therefore, it takes the second moon 28729, or 8,323, days to orbit the planet 29 times. Since it takes the first moon 7,308 days and the second moon 8,323 days, it takes the second moon 8,323 - 7,308, or 1,015, more days than it takes the first moon to orbit the planet 29 times.

Q7

24

The correct answer is 24. The equation $\frac{24x}{ny} = 4$ can be rewritten as $\frac{24}{n}\frac{x}{y} = 4$. It's given that $\frac{x}{y} = 4$. Substituting 4 for $\frac{x}{y}$ in the equation $\frac{24}{n}\frac{x}{y} = 4$ yields $\frac{24}{n}4 = 4$. Multiplying both sides of this equation by n yields $244 = 4n$. Dividing both sides of this equation by 4 yields $24 = n$. Therefore, the value of n is 24.

Q8

A

A. $\frac{4}{3}w$

Choice A is correct. It's given that rectangle A has length 15 and width w. Therefore, the length-to-width ratio of rectangle A is 15 to w. It's also given that rectangle B has length 20 and the same length-to-width ratio as rectangle A. Let x represent the width of rectangle B. The proportion $\frac{15}{w} = \frac{20}{x}$ can be used to solve for x in terms of w. Multiplying both sides of this equation by x yields $\frac{15x}{w} = 20$, and then multiplying both sides of this equation by w yields $15x = 20w$. Dividing both sides of this equation by 15 yields $x = \frac{20w}{15}$. Simplifying this fraction yields $x = \frac{4}{3}w$.

Choices B and D are incorrect and may result from interpreting the difference in the lengths of rectangle A and rectangle B as equivalent to the difference in the widths of rectangle A and rectangle B. Choice C is incorrect and may result from using a length-to-width ratio of w to 15, instead of 15 to w.

Q9

A

A. 25 meters per second

Choice A is correct. Since 1 kilometer is equal to 1,000 meters, it follows that 90 kilometers is equal to $90(1{,}000) = 90{,}000$ meters. Since 1 hour is equal to 60 minutes and 1 minute is equal to 60 seconds, it follows that 1 hour is equal to $60(60) = 3{,}600$ seconds. Now $\frac{90 \text{ kilometers}}{1 \text{ hour}}$ is equal to $\frac{90{,}000 \text{ meters}}{3{,}600 \text{ seconds}}$, which reduces to $\frac{25 \text{ meters}}{1 \text{ second}}$ or 25 meters per second.

Choices B, C, and D are incorrect and may result from conceptual or calculation errors.

Q10 4

The correct answer is 4.44. The selling price, in dollars per cubic inch, is found by multiplying the density, in ounces per cubic inch, by the unit price, in dollars per ounce: $\left(\frac{0.555 \text{ ounce}}{1 \text{ cubic inch}}\right)\left(\frac{\$8.00}{1 \text{ ounce}}\right)$ yields $\frac{\$4.44}{1 \text{ cubic inch}}$. Thus, the selling price, in dollars per cubic inch, is 4.44.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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